

Ryan Liu

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EDUCATION

Northwestern University, McCormick School of Engineering | Evanston, IL Expected Jun 2029

- B.S. Mechanical Engineering | Major GPA: 3.85/4.00 | Cumulative GPA: 3.36/4.00 | Dean's List, Spring 2026
- Relevant Coursework: Mechanical Design & Manufacturing, Properties of Materials, Engineering Analysis, Smart Electronics in Design; Upcoming Fall 2026: Design of Elements, Computer Integrated Manufacturing, Electronics Design

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, NX, Fusion 360, Inventor, FEA, GD&T, DFM, assemblies, engineering drafting
Manufacturing & Prototyping: CAM, CNC milling, manual milling/turning, FDM/SLS 3D printing, rapid prototyping
Electronics & Programming: Circuit prototyping, soldering, CAN, Arduino, PID control, MATLAB, Git

EXPERIENCE

Northwestern Formula Racing | Chassis Subteam Member, Steering & Drivetrain | Evanston, IL Sep 2025 - Present

- Developing the 2027 steering system with carbon-fiber tubing under chassis and driver-clearance constraints, targeting a 50% mass reduction over the previous steel design while meeting torque and stiffness requirements
- Performed static and fatigue FEA on a concentric spline-to-sprocket differential adapter, reducing mass by 35% while meeting strength and life-cycle requirements
- Leading R&D of carbon-fiber tube insert bonding for steering and suspension applications
- Defined steering load cases by calculating steering-wheel torque from suspension geometry and rack ratio
- Designed and CNC-machined the differential adapter from 7075 aluminum on a Haas VF-2
- Programmed a 6-hour, 2-setup, multi-tool CAM process in Fusion 360 and version-controlled CAD files using Git

Crescent Coyotes - FIRST Robotics Team 610 | Design & Electrical Lead, Robot Driver | Toronto, ON Sep 2022 - Jun 2025

- Directed the design and assembly of two 150-lb competition robots, optimizing complex subsystems (swerve drivetrains, elevators, intakes, and manipulators) for packaging, manufacturability, serviceability, and rapid iteration
- Led electrical integration, wiring, and troubleshooting across power distribution, CAN networks, motors, sensors, cameras, and Orange Pi/Jetson Nano computers
- Designed sensor mounts and electronics enclosures for computer-vision systems, ensuring reliable state detection, unobstructed target visibility, and secure cable routing
- Developed 10 CAD/CAM tutorials for 60+ students and partnered with school administrators to secure \$30,000 in hardware upgrades and access to a 7,200 sq. ft. practice facility
- Drove the team to 4 event finals and 9 awards, improving district ranking from 23rd to 10th among 130 teams in Ontario

Discobots Canada - VEX Robotics Team 1140Z | Technical Lead, Robot Driver | Brampton, ON May 2018 - Sep 2022

- Directed the design and assembly of 15-lb competition robots integrating drivetrains, lifts, intakes, flywheels, and pneumatic manipulators within an 18-in. cube size constraint
- Programmed autonomous routines with PID motion control using sensor feedback (color, line, and object detection) to maximize scoring efficiency
- Led the team to 2 VEX Worlds Division Finals and 20+ provincial awards

PROJECTS

3D-Printed Bicycle Brake Calipers

- Designed rear bike brake calipers in NX to meet ISO braking requirements, using FEA to guide topology optimization and evaluate stress concentrations, deflection, load paths, and fatigue life
- Defined critical load cases by calculating required cable tension and brake pad normal force based on system mass, target stopping distance, lever geometry, and rider input force
- Fabricated and iterated on Nylon 12 (SLS) prototypes through testing, reducing braking distance by 15% with 4% added mass

Interactive LED Sensory Panel

- Created CAD models and CAM toolpaths in Fusion 360 for a CNC-pocketed HDPE panel integrating reflective foil, 3D-printed structural supports, LED mounts, and a custom low-profile tactile button
- Wired LED/button electronics and programmed Arduino lighting behaviors to engage students with intellectual and developmental disabilities

Waist-Mounted DSLR Support Arm

- Developed an accessible waist-mounted DSLR support system in Fusion 360 to redistribute camera weight for photographers with limited upper-body mobility
- Modified a COTS camera arm with a custom-machined hex shaft, 3D-printed arm bracket, and laser-cut support panels